SPE-229779-MS

# Enhanced Well Intervention Efficiency Using Digital Slickline: A Comparative Study Between Digital and Conventional Slickline Operations

Ashutosh Kumar, TAQA, Abu Dhabi, United Arab Emirates; Raid Bukahmseen, TAQA, Dammam, Kingdom of Saudi Arabia; Lyes Ben Moussa and Ismail P Mamu, TAQA, Abu Dhabi, United Arab Emirates

Copyright 2025, Society of Petroleum Engineers DOI 10.2118/229779-MS

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

## Abstract

Deployment of digital slickline across multiple wells with various complexities, including mechanical intervention such as setting and retrieving plugs and gauge hanger, jet pump changeout, manipulation of sliding side door (SSD), and well integrity check. Real-time downhole data acquisition, including depth correlation, tension, pressure, and temperature measurement, was integrated into operations to enhance surface-to-downhole communication, which is a crucial part of many common well intervention operations. This paper will introduce digital slickline technology and describe how it differs from conventional slickline and opening a wider range of technology applications and implementation options. The paper compares the new technology digital slickline with conventional slickline in terms of engineering design, operational mechanism, and requirements. The digital slickline offers a number of features, which will be covered in the paper, that make this technology especially useful in many special operational circumstances. This paper includes the pre-job planning with digital modelling execution of the job with conventional and digital slickline tools, and post-job analysis using statistical operational metrics data, such as rig up/down time, intervention duration, runs counts, success rates, and nonproductive time (NPT) were collected and analysed.

The study provides for an analysis and statistical review of 50 well interventions with slickline in terms of several key performance indicators KPIs such as nonproductive time (NPT), operating hours, operational cost, and risk of real-time monitoring of well intervention with slickline. Those 50 well interventions are evenly divided between conventional slickline and deployments of digital slickline. The field deployment of digital slickline resulted in an increase in well intervention scenarios and possibilities compared to conventional slickline, in addition to a reduction of 25% to 35 % total intervention time. This was mainly due to the enhanced slickline engineering with real-time monitoring. Deployment of the digital slickline also resulted in a noticeable 20 % reduction in operational cost and a reduction of NPT hours of 40%. Real time monitoring enable depth control and verification of downhole tool status which leads to a reduction of 20 % in repeated runs, study also revealed that real time tension monitoring reduced the risk of tool stuck and line break which result in improvement of operational safety, Overall, the digital slickline was found to be not only an expansion of the slickline technology deployment reach but also a noticeable value-adding in terms

Downloaded from https://onepetro.org/SPEADIP/25ADIP/conference/25ADIP by Dmitry Kruglov on 09 November 2025

of operations efficiency and data quality. The paper provides the readers with a thorough description of a high-tear technology innovation in slickline. The novel integration of real time data to transform slickline from a purely mechanical intervention system to an intelligent intervention system, which introduces the ability to widely expand the range of digital slickline deployment possibilities and reduce the operational cost while improving performance efficiency in modern well intervention practices.

## Introduction

The oil and gas industry continuously seeks innovative technologies that enhance operational efficiency, while reducing costs and improving safety during well intervention operations. The conventional slickline coast effectiveness and reliability have been limited due to the lack of real-time downhole data and precise depth control. This limitation has often necessitated the necessitated of multiple conveyance systems and increased the operational complexity for comprehensive well intervention programs.

Recent technological advancements have introduced digital slickline systems that fill the gap between conventional mechanical slickline operations and electric line services. Digital slickline systems is the integration of real-time data acquisition, bi-directional telemetry, and enhanced depth correlation capabilities while maintaining the fundamental advantages of conventional slickline usage. This technology addressed critical industry needs for the improvement of operational efficiency, reduction of non-productive time, and enhanced safety in well intervention activities.

This paper presents a detailed, comprehensive study of digital slickline versus conventional slickline based on the operational data from 50 well intervention activities and the technology upgradation in the digital slickline. This study is based on the key performance indicators, including well intervention duration, operational costs, non-productive time, and safety metrics, to quantify the benefits of digital slickline technology adoption.

## Digital Slickline Technology Overview

Digital slickline presents a revolutionary advancement in well intervention technology, combining the reliability of conventional slickline with sophisticated real-time data abilities. The core innovation of digital slickline lies in the integration of a polymeric coating around conventional slickline wire, which enables bi-directional digital communication while maintaining mechanical strength and operational versatility. The technology includes several key advantages over conventional approaches of slickline. The real-time data acquisition provides continuous monitoring of downhole conditions like pressure, temperature, depth correlation, and tool status. This ability of digital slickline eliminates the risk of uncertainty associated with traditional memory-based systems and enables immediate operational adjustments based on actual downhole conditions.

Enhanced depth correlation shows another significant advancement in slickline, with digital system slickline achieving electric-line accuracy on conventional slickline setups. This condition enables more accurate tool placement, reduces repeated runs, and improves overall operational efficiency. The continuous depth correlation ability is valuable in complex well intervention, where precise positioning is critical for successful intervention operations. The integrated approach of digital slickline eliminates the need for multiple conveyance systems, as a single setup can handle both mechanical interventions and data acquisition operations. This consolidation reduces equipment requirements, personnel needs, and logistical complexity while improving operational efficiency and safety. Modern digital slickline systems also incorporate advanced safety features, including real-time tension monitoring, shock measurement, and automated release mechanisms. These capabilities provide enhanced operational safety by detecting potentially hazardous conditions before they result in equipment damage or personnel injury.

Digital slickline systems contain three primary components: surface equipment, the specialized cable system, and downhole assemblies. The surface equipment includes the Communication data processing

Downloaded from https://onepetro.org/SPEADIP/25ADIP/conference/25ADIP by Dmitry Kruglov on 09 November 2025

panel, which serves as the central control and data acquisition hub. This system processes real-time data from downhole sensors and provides operators with continuous feedback on operational parameters, including depth, tension, speed, and tool status. The Communication antenna enables contactless communication with the downhole assembly through the coated cable system. This wireless interface eliminates the need for slip rings and complex electrical connections while providing robust data transmission capabilities. The intelligent Sheave provides accurate depth, tension, and speed measurement, effectively replacing conventional measuring heads and improving measurement accuracy. The digital slickline cable represents the core innovation, featuring a precision polymeric coating applied to high-strength slickline wire. The coating serves dual purposes, reducing friction and abrasion while enabling bi-directional digital communication. The cable maintains the mechanical properties of conventional slickline while adding sophisticated data transmission abilities. The typical specification of digital slickline wire includes a 0.160-inch outer diameter with a 3240-pound breaking load and operational capabilities up to 15000 psi and 350°F. Downhole components include the cable head, communication head (CH), and power processing unit (PPU). The cable head provides a secure mechanical connection to the tool string while maintaining electrical continuity for data transmission. The communication head manages bi-directional data flow between surface and downhole systems, while the power processing unit supplies electrical power to downhole sensors and tools. The integrated sensor suite within the PPU includes pressure and temperature sensors, casing collar locator, three-axis accelerometer, and tension measurement capabilities. This comprehensive sensor package provides real-time feedback on downhole conditions and tool performance, enabling informed operational decisions and enhanced safety monitoring.

In Fig.1, the complete setup of the digital slickline is shown with the important components of the digital slickline which consist of a winch, data processing panel (DP), a central junction box, communication antenna and stuffing box, sheave, digital slickline wire, cable head or rope socket, communication head, shear pin release tool.

The digital slickline winch is the next generation, full electric, easy-to-operate, silent, and has reduced maintenance for the winch unit. Winch supports cased hole wireline services with the smallest footprint, reduction of unit cost, and minimal crew requirements. The single lift design cuts non-productive time and costs for unit change-out. Winch is based on a computer-controlled all-electric drive, which instantaneous response gives extremely fast stop capability, driven by operator-defined parameters. The winch double drum configuration supports standard slickline wires 0,108 to 0,160", digital slickline, 3/16 Ultra Line, and 7/32 cables. The winch unit features a compact, ergonomic cabin with simplified and standard manual controls, automated spooling, and digital and analogue displays providing depth, tension, speed, and visualization data. It operates at constant speeds ranging from 0.2 to 1,200ft/min with constant tension control, supports remote operation from any well or office location, and enables real-time communication with the office for operational support. This system offers a reduced deck footprint and unit costs, minimizes crew requirements, and supports ultra-slow logging speeds of 0.1ft/min. Additional benefits include automated speed management, programmable logging operations, automatic jarring, job performance monitoring, line tension prediction per run, and comprehensive monitoring of line history and fatigue.

Data Processing Panel (DP) is an industrial computer that processes depth, tension, and data to and from downhole tools. The DP is the central component for a real-time digital slickline system and sets the new Industry standard for ultra-precision, safety, and control for the latest innovation of electronically controlled E-Winch units, another step change in operational efficiency. The system features bi-directional communication between the DP panel and digital slickline downhole tools, with digital and analogue displays showing depth, speed, and tension. It includes certified models suitable for rig-safe or zone-2 areas and provides visualization of the tool string position within the well. Additional capabilities include job performance monitoring, programmable alarms and shutdown limits, manual winch control via joystick or screen buttons, and automated management of speed, tension, jarring, and logging, all while maintaining constant speed and tension.

Downloaded from https://onepetro.org/SPEADIP/25ADIP/conference/25ADIP by Dmitry Kruglov on 09 November 2025

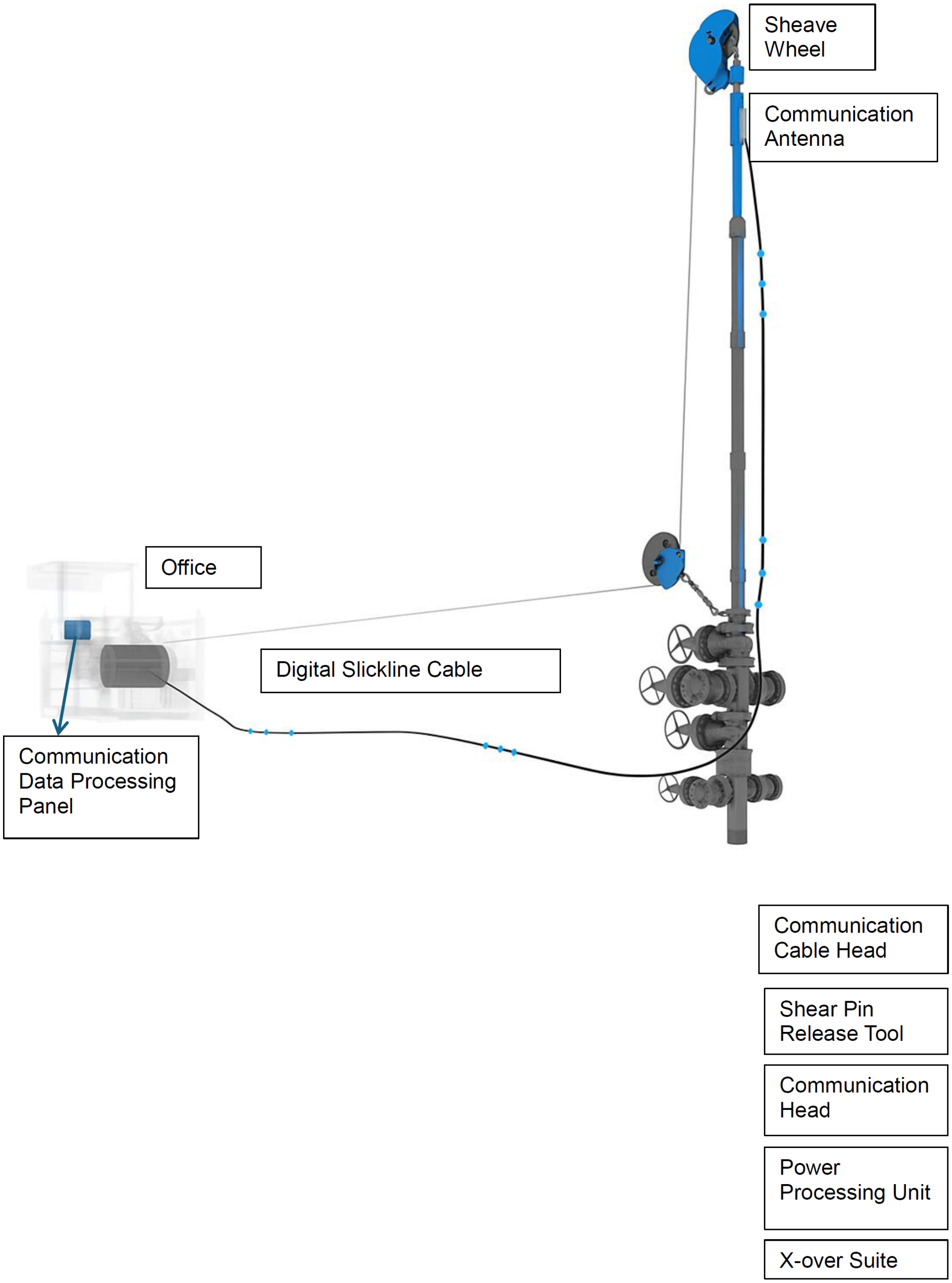


**Figure 1—Digital Slickline set up**

The digital slickline central junction box is one of the available central boxes of the communication system. Central junction box used to provide both power and communication to the data processing panel, antenna, and sheave, but also to communicate with the logging computer with log or office. The box fitted with connectors to fit the cable assemblies to all equipment items.

The antenna used for the data communication with the digital slickline cable and transmits data to and from the downhole tools. The antenna incorporates a communication box and lubricator section specially

Downloaded from https://onepetro.org/SPEADIP/25ADIP/conference/25ADIP by Dmitry Kruglov on 09 November 2025

designed for the digital slickline system. The antenna module integrated within a standard lubricator section connected to a standard slickline stuffing box that also supports the top sheave that measures depth, speed, and tension.

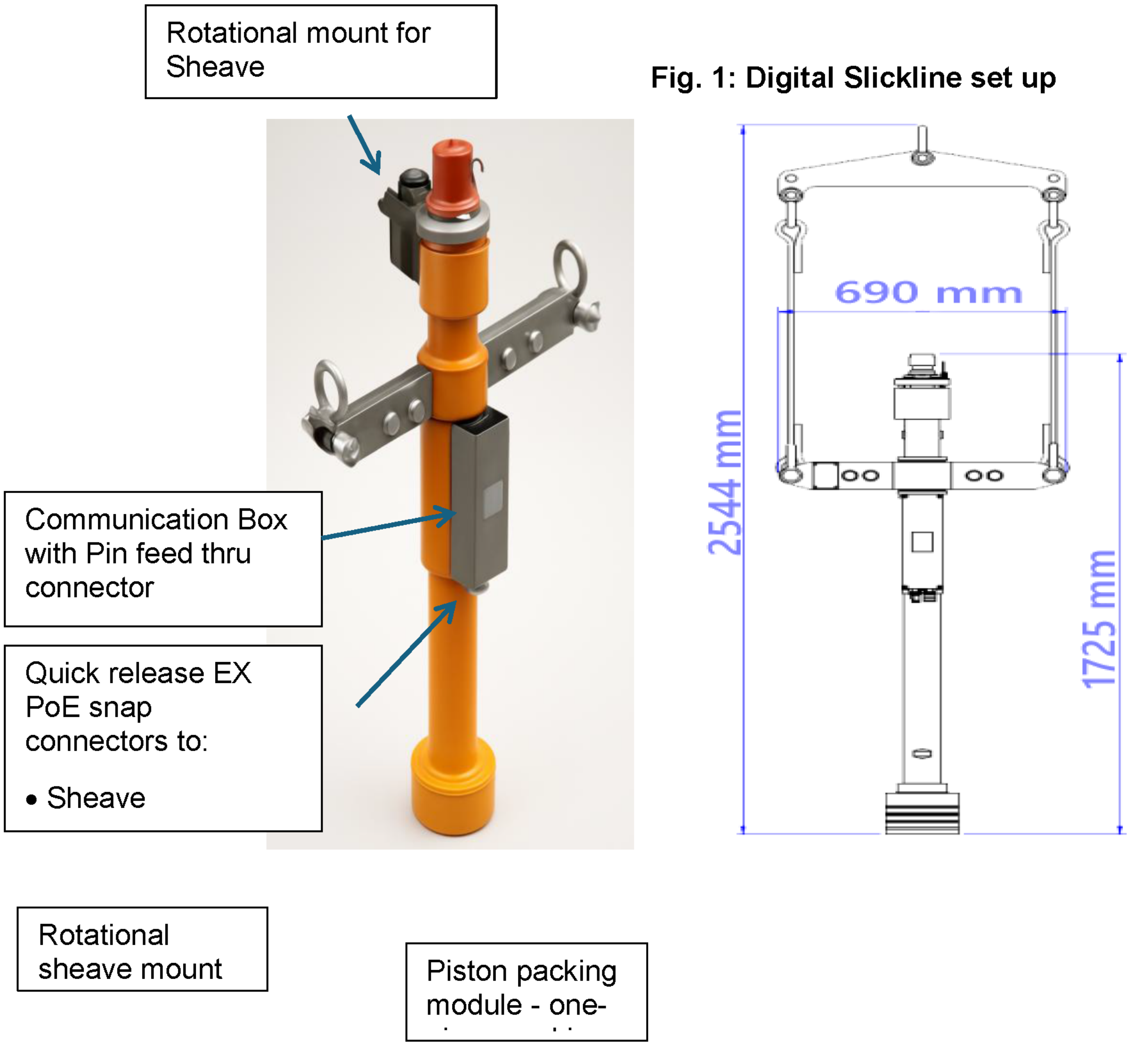


**Figure 2—Antenna Assembly**

The stuffing box's innovative design seals on the coated digital slickline wire against the well pressure. The sealing element is designed with a longer, large cross-sectional area, which provides an environmentally friendly seal and cleaning action at well intervention operations. The stuffing box is designed to pack off all sizes of slickline wire and coated cables to a maximum of 0.160” diameter using a 20” sheave. The bottom connection can be suited to standard pressure control equipment (PCE) size. The features of the stuffing box include a one-piece packing stick with a large cross-sectional seal area, a quick-connect block for the sheave, and a lockable, free-rotational sheave mount. The system offers oil or grease injection through a dedicated piston port, as well as a glycol or chemical injection port. It is equipped with a grease nipple for preventive maintenance of the roller bearings in the rotating sheave mount and an integrated seal sub and collar assembly with a ball safety check valve and a blow-out plug.

Downloaded from https://onepetro.org/SPEADIP/25ADIP/conference/25ADIP by Dmitry Kruglov on 09 November 2025

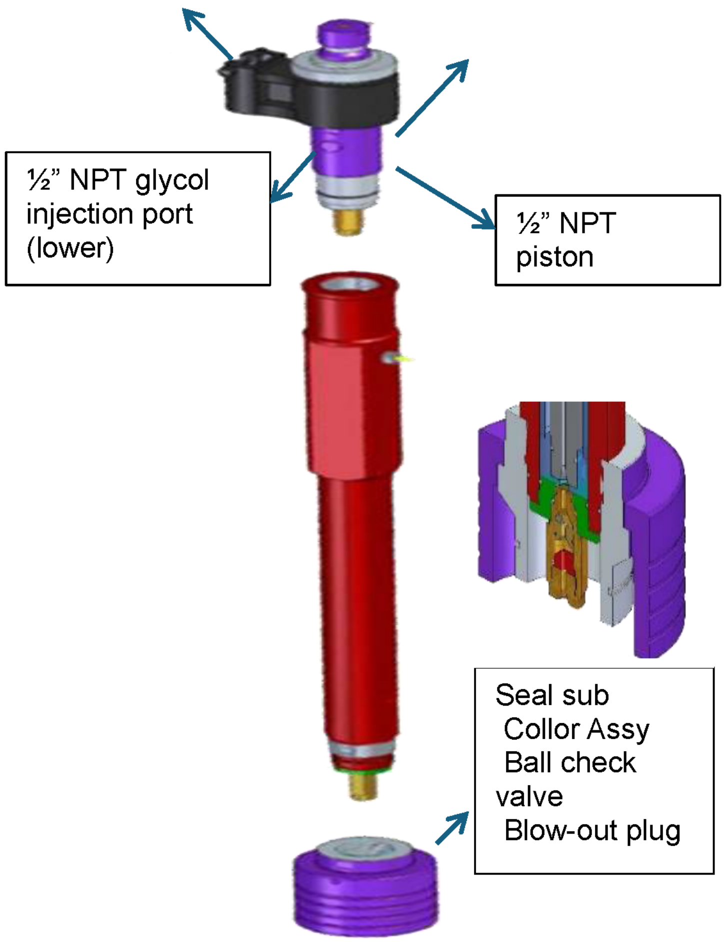


**Figure 3—Stuffing box assembly**

The sheave is an intelligent sheave assembly that guides the wire into the well and, at the same time, is integrated with measuring sensors for depth, speed, and tension. The sheave is mounted on top of the antenna with a stuffing box. The sheave is equipped with explosion-proof electronics to communicate depth, speed, and tension data to the antenna that is connected to the data processing panel. By using the sheave wheel, there is no need for a conventional measuring head system that is normally installed on the winch unit.

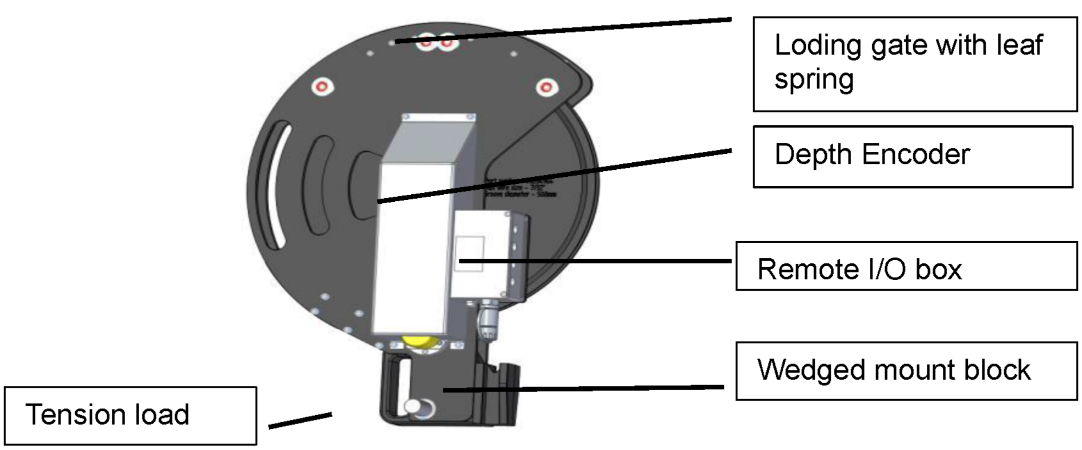


**Figure 4—Sheave assembly**

Downloaded from https://onepetro.org/SPEADIP/25ADIP/conference/25ADIP by Dmitry Kruglov on 09 November 2025

## Comparative Study Methodology

The comparative analysis of conventional and digital slickline was done in 50 well intervention operations performed across multiple field locations, with an equal portion of intervention, 25 operations performed using conventional slickline and 25 using digital slickline technology. The study ensured similar well conditions, intervention complexity, and operational objectives to enable a fair performance comparison. Well selection criteria included similar completion, comparable depth profiles, and equivalent intervention scopes to minimize variable conflicts that could influence comparative results. Operations include mechanical interventions such as plug setting and retrieval, gauge hanger manipulation, jet pump changeouts, and sliding side door operations. Each intervention was documented with detailed operational metrics, including rig-up and rig-down times, intervention duration, number of runs, success rates, and non-productive time incidents. Intervention time was measured from initial equipment mobilization to final demobilization, including all operational phases. Operational costs consist of personnel, equipment, and support service expenses that directly contribute to the intervention. Non-productive time was defined as any period during which intervention progress was halted due to equipment issues, operational challenges, or safety concerns. Quality control measures included independent verification of operational data, standardized reporting protocols, and peer review of collected information. Statistical analysis methods employed confidence interval calculations and variance analysis to ensure the reliability and significance of comparative results. The study also consists of qualitative assessments of operational complexity, crew confidence levels, and post-operation feedback from field personnel. These subjective measures provided additional context for interpreting quantitative performance data and understanding the practical implications of technology adoption.

## Field Testing and Implementation

Field implementation of digital slickline technology required crew training and adaptation of operational procedures. Initial deployments of digital slickline are focused on the establishment of the baseline performance metrics and on identifying the optimization opportunities specific to digital system abilities. The learning and adoption of new technology was carefully monitored and documented to understand its impact on operational efficiency. Pre-job planning for digital slickline operations incorporated enhanced modelling capabilities enabled by real-time data feedback. Digital modelling execution of the intervention allowed for more accurate operational planning and improvement in contingency preparation; the ability to monitor real-time downhole conditions enabled a dynamic operational adjustment in between the operations that were not possible with conventional systems. Operational execution revealed significant advantages in tool deployment and retrieval operations. Real-time depth correlation eliminates the uncertainty in tool positioning while continuous tension monitoring provides enhanced safety oversight. The ability to verify the tool status before pulling out of the hole reduces repeated runs and improves operational efficiency. Integration challenge of digital slickline includes the adaptation of existing operational procedures to leverage digital system capabilities fully. Personnel training requirements include both technical operation and interpretation of real-time data streams. Initial implementation phases required close technical support to optimize system utilization and resolve operational questions. Field testing also validates the quality of digital slickline under challenging operational conditions. The polymer-coated cable demonstrated excellent resistance to abrasion and chemical exposure while maintaining data transmission integrity. Downhole electronic components provide reliable data under high-pressure and temperature conditions, confirming system suitability for demanding intervention environments.

Downloaded from https://onepetro.org/SPEADIP/25ADIP/conference/25ADIP by Dmitry Kruglov on 09 November 2025

## Results and Performance Analysis

The comparative study resulted in significant performance improvement across all measured parameters when the digital slickline technology was deployed in the field total intervention time shows a reduction of 25% to 35% with an average of 30% improvement across all operations. This time reduction was primarily attributed to enhanced operational efficiency enabled by real-time monitoring and improved depth correlation accuracy. The operational cost of digital slickline analysis revealed a consistent 20% reduction in total intervention expenses when using digital slickline systems. Cost savings are derived from multiple sources, including reduced personnel requirements, eliminated equipment changeouts, and decreased non-productive time. The single system approach of digital slickline eliminated the need for multiple conveyance setups, resulting in significant low logistical cost and non-productive time incidents decreased by 40% with digital slickline deployment this improvement reflects enhanced operational safety through real-time monitoring abilities and reduced equipment related issues the ability to monitor continuously downhole conditions enabled active identification and resolution of potential intervention problems before they resulted in operational delays. Repeat runs are decreased by 20% due to improved depth accuracy and real-time tool status verification. Digital system enabled operators to confirm successful tool deployment and operation before retrieving equipment, eliminating uncertainty-driven repeated interventions. This improvement contributed to overall time and cost reductions. Success rates improved across all operation types, with digital slickline achieving 98% first-time success compared to 89% for conventional slickline. The enhanced success rate reflects improved operational quality and real-time feedback abilities that enabled immediate corrective actions when required. Statistical analysis confirmed the significance of performance improvements, with confidence intervals indicating reliable and repeatable benefits from digital technology adoption. Variance analysis demonstrated consistent improvement patterns across different well types and operational conditions, validating the universal applicability of digital slickline advantages.


**Figure 5—Performance comparison between conventional and digital slickline operations showing significant improvements across key operational metrics**

Downloaded from https://onepetro.org/SPEADIP/25ADIP/conference/25ADIP by Dmitry Kruglov on 09 November 2025

## Case Studies and Applications

### Case Study 1: SSD (Sliding Sleeve Door) Operations

A case study involved locating and manipulation of a SSD to open the zone for production where the conventional slickline is unable to locate the SSD due to wax buildup and depth uncertainty the digital slickline were deployed with real-time casing collar locator abilities successfully identified the SSD at 9250 feet significantly different from the 9500 feet depth indicated on completion drawing the digital system real-time pressure and temperature monitoring enabled optimization of manipulation of SSD 0perations by providing immediate feedback on jar performance and tool engagement. Successful SSD open was confirmed through downhole temperature and pressure response changes, eliminating uncertainty about operation completion. The total operation time was reduced to 4 hours compared to a full day of unsuccessful conventional slickline attempts.

As shown in the graph 3, the blue line indicate the jar action to open the SSD, at the point where the SSD opens the yellow line in the graph which represent the pressure shows the decline and there is raise in temperature graph both the indication leads to the conclusion that the SSD is open, there is no need for further runs to confirm whether the SSD is open or closed.

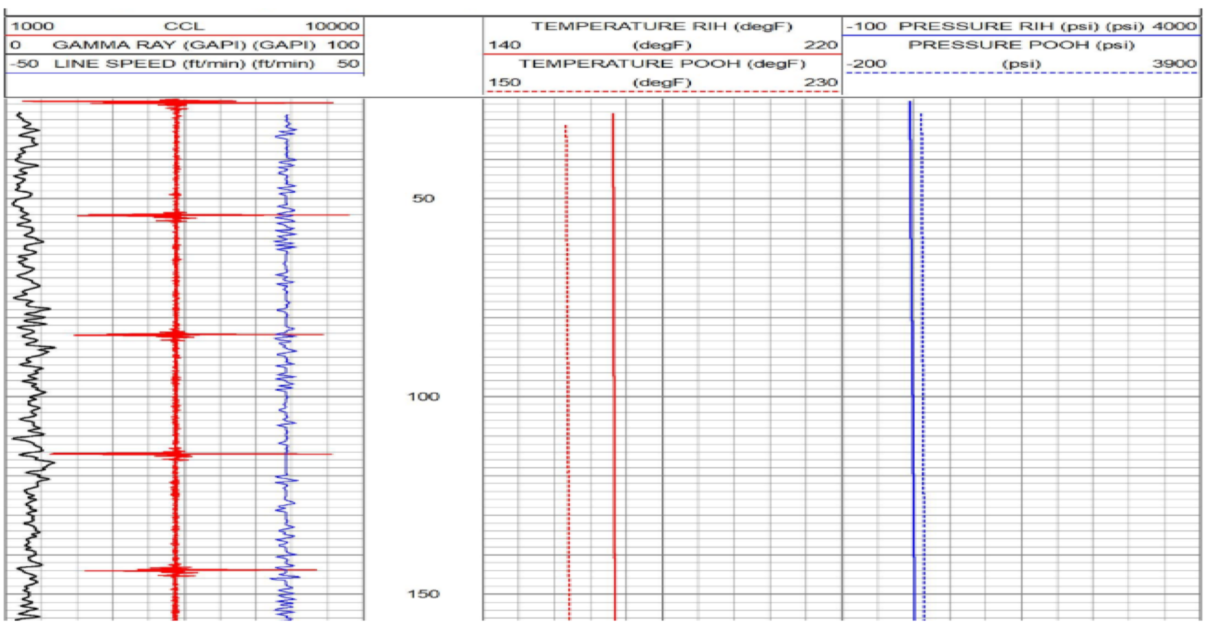


**Figure 6—CCL, Line tension, Line Speed, Temperature and pressure data displayed at the surface in real time, which enables to locate the SSD.**

Downloaded from https://onepetro.org/SPEADIP/25ADIP/conference/25ADIP by Dmitry Kruglov on 09 November 2025

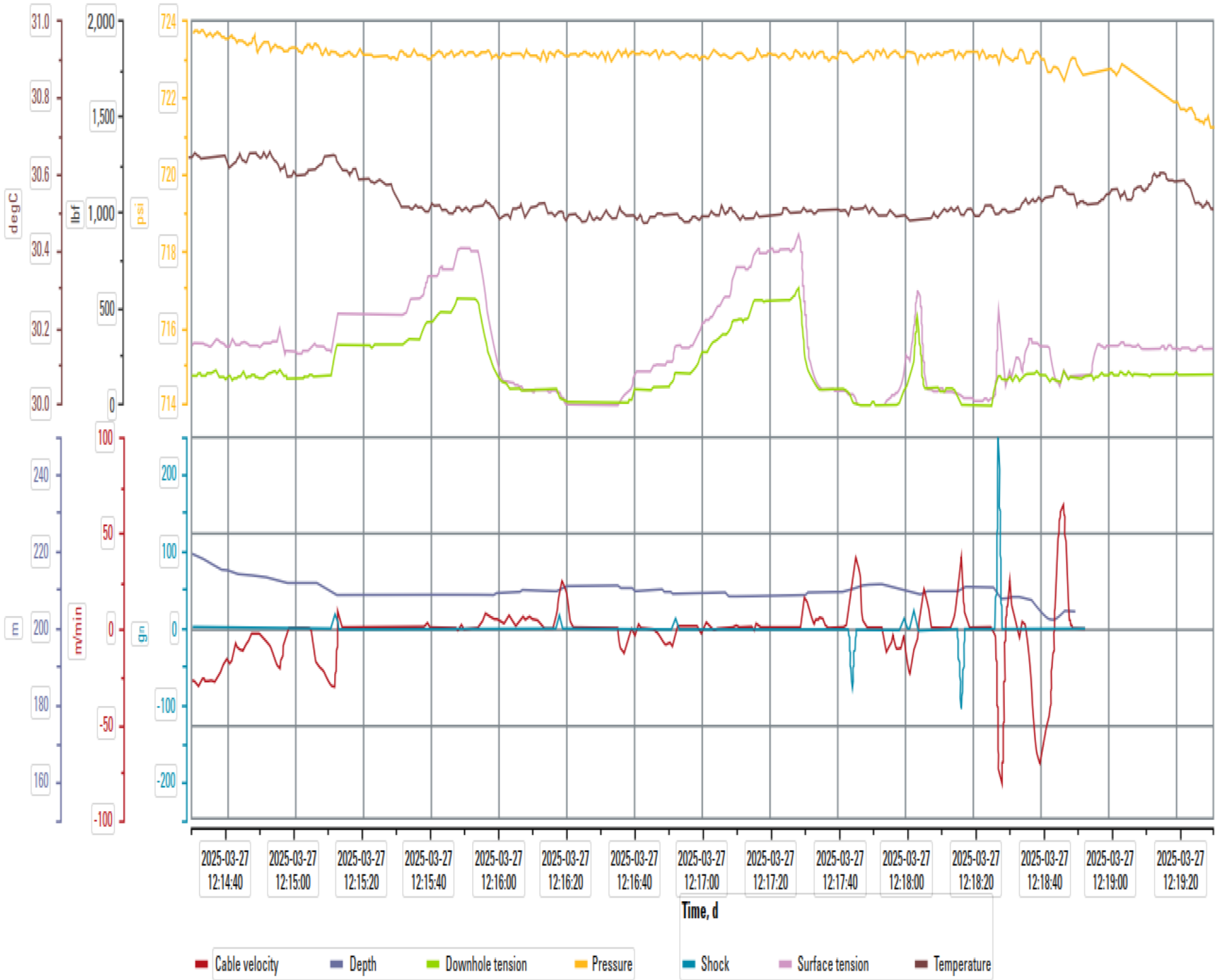


**Figure 7—Monitoring of the jarring operation to open SSD.**

## Case Study 2: Production Logging Integration

A case study involved the jet pump change out and production logging test in which slickline is planned to perform the jet pump retrieval gauge cutter run SSD manipulation and electric line to perform the production logging, later on the digital slickline were deployed to the well with production logging, operation demonstrated the technology versatility in combining mechanical and diagnostic services real-time production profiling eliminated the need for separate electric line deployments while providing comprehensive well performance data. The system enabled flowing surveys with optimized flow rates based on real-time head tension monitoring, maximizing data quality while maintaining operational safety. The case demonstrated digital slickline's capability to perform traditional electric line services with slickline equipment simplicity and cost-effectiveness by eliminating the waiting hours for both the slickline and electric line units. Real-time data availability enabled immediate interpretation and operational decisions, eliminating delays associated with post-operation data processing and analysis.

Downloaded from https://onepetro.org/SPEADIP/25ADIP/conference/25ADIP by Dmitry Kruglov on 09 November 2025

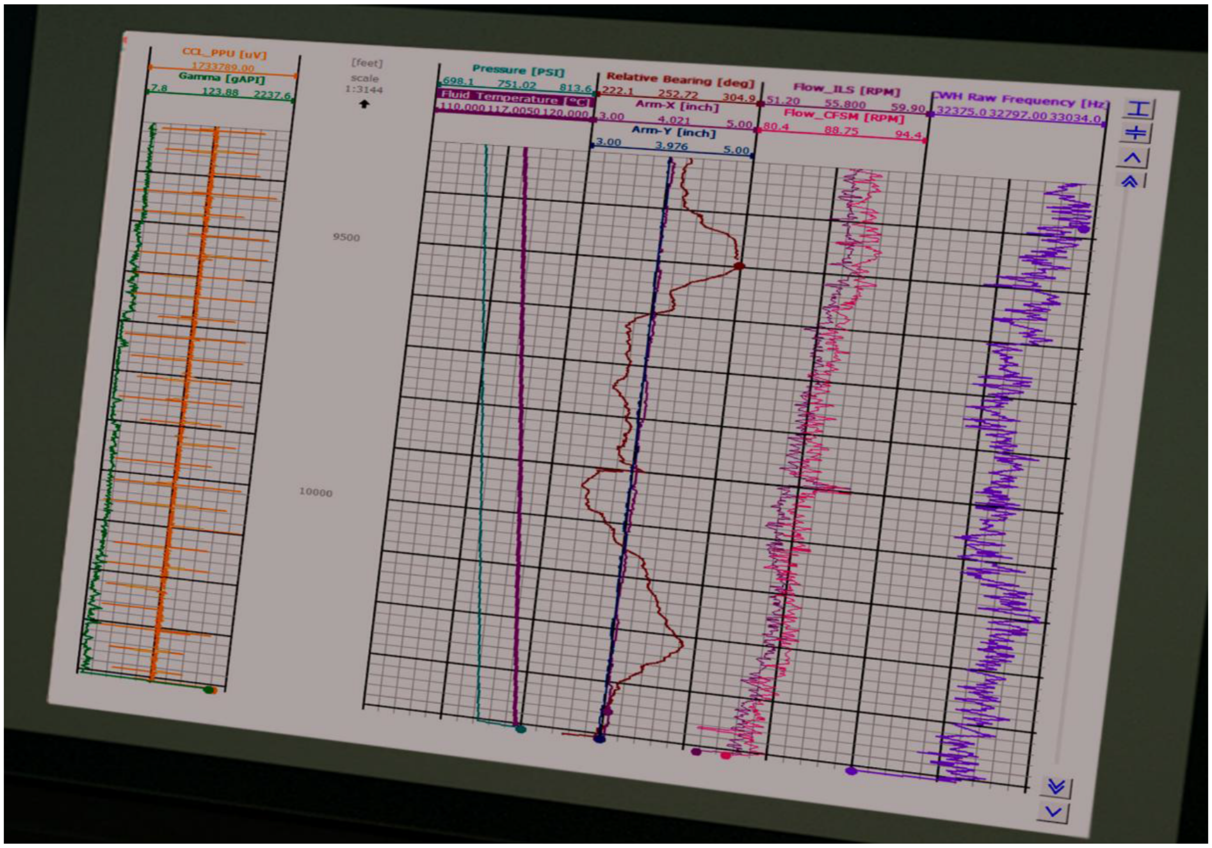


**Figure 8—Showing the real-time data of production logging**

## Economic Impact Analysis

The economic benefits of digital slickline technology extend beyond direct operational cost reductions to encompass broader value creation across well intervention programs personnel cost reductions average 30% due to integrated system capabilities that eliminate the need for specialized crews and product line for different intervention phases equipment cost savings of 25% result from single system deployments that replace multiple conventional setups rig up/down time savings represent the most significant economic benefit with 30% reductions in overall intervention duration translating to substantial daily rate savings. In offshore environment where rig rates can exceed $30000 per daytime savings of 120 hours per multi-well campaign generate cost avoidance exceeding $150000 these savings scale proportionally with campaign size and rig day rates non-productive time cost reduction provide additional economic benefits through improved operational reliability the 40% reduction in NPT incidents translates to approximately $10000 per well in avoided costs based on typical well intervention economics over large intervention programs, these savings accumulate to significant program level benefits.

Opportunity cost benefits include reduced production deferrals due to shorter intervention durations, wells returned to production 1-2 days earlier, generating additional revenue that offsets technology implementation costs for high-value production wells. This production acceleration can justify digital slickline adoption independently of direct cost savings. Technology lifecycle costs remain competitive with conventional alternatives due to the durability and reliability of digital system components. The polymer-coated cable demonstrates extended service life compared to conventional slickline wire, while electronic components prove robust under field conditions, maintenance requirements remain minimal, ensuring a favourable total cost of ownership.

Downloaded from https://onepetro.org/SPEADIP/25ADIP/conference/25ADIP by Dmitry Kruglov on 09 November 2025

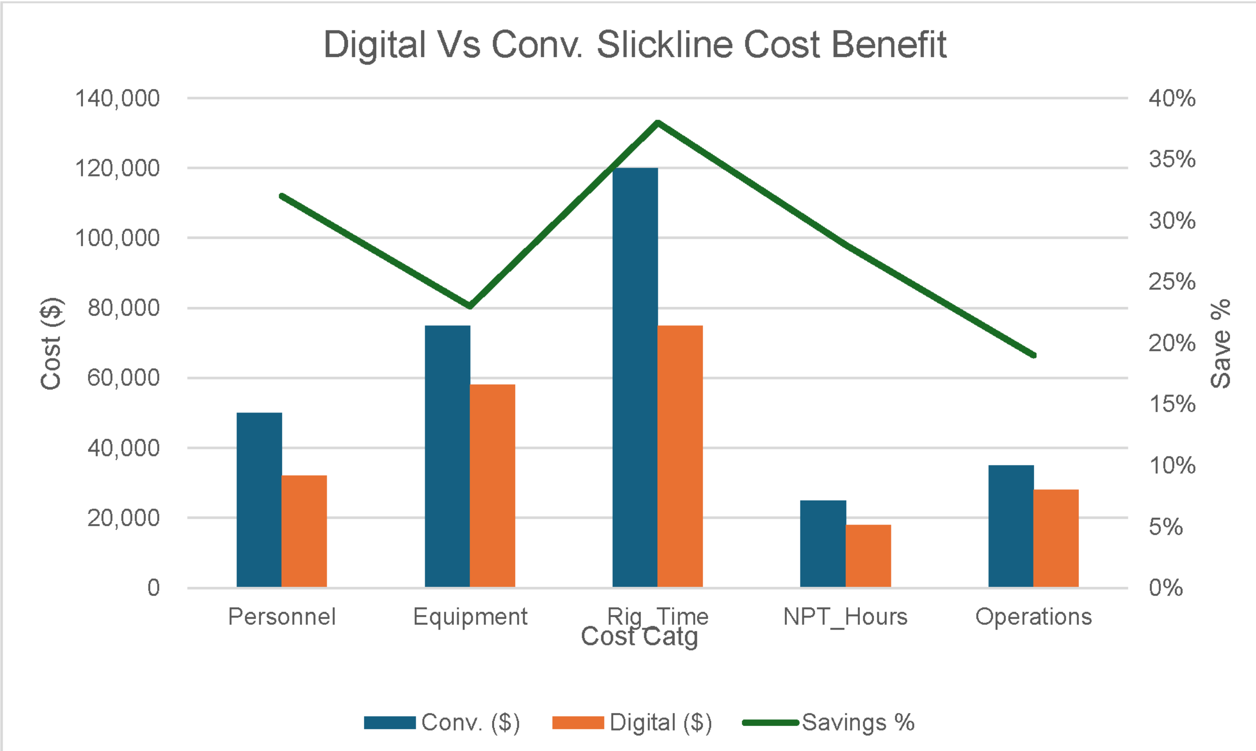


**Figure 9—Economic comparison between conventional and digital slickline operations showing cost savings across multiple operational categories**

## Safety and Risk Assessment

Digital slickline technology delivers substantial safety improvements through enhanced monitoring abilities and active risk identification real-time tension monitoring provides continuous oversight of mechanical loading conditions enabling early detection of potential tool sticking or cable stress issues this ability prevents equipment failures that could result in personnel injury or environmental incidents shock measurement systems monitor jarring operations and tool impacts providing quantitative feedback on operational forces this data enables optimization of mechanical operations while preventing excessive loading that could damage equipment or compromise wellbore integrity. Historical data analysis supports predictive maintenance programs and operational parameter optimization, and the elimination of equipment changeouts reduces personnel exposure to lifting operations and pressure control equipment manipulation. Single-system operation minimizes handling requirements and reduces opportunities for human error during equipment configuration changes. Simplified logistics reduce transportation requirements and associated safety risks. Real-time operational monitoring enables immediate response to developing safety concerns, preventing minor issues from escalating to significant incidents. Continuous data streams provide early warning of abnormal conditions, supporting proactive intervention and risk mitigation strategies. Personnel competency benefits include reduced complexity in operational procedures and standardized equipment interfaces. Digital systems provide consistent operational protocols across different intervention types, reducing training requirements and minimizing procedural errors. Enhanced data availability supports improved decision-making and reduces operational uncertainty.

## Future Developments and Trends

The evolution of digital slickline technology continues with advanced abilities in development and field-testing phases, enhanced telemetry systems promise higher data transmission rates and improved reliability under challenging wellbore conditions. Increased bandwidth capabilities will enable more sophisticated downhole sensors and real-time video transmission for visual well inspection applications.

Downloaded from https://onepetro.org/SPEADIP/25ADIP/conference/25ADIP by Dmitry Kruglov on 09 November 2025

Autonomous operational capabilities represent the next frontier in digital slickline evolution. Automated conveyance systems demonstrate the potential for consistent operational execution with reduced human intervention. These systems show promise for standardizing operational procedures and eliminating human variability in intervention execution. Integration with artificial intelligence and machine learning algorithms will enable predictive operational optimization and automated decision-making support. Real-time data analysis abilities will identify optimal operational parameters and predict equipment performance issues before they impact operations. Advanced sensor integration will expand diagnostic capabilities to include formation evaluation, production analysis, and well integrity assessment functions that traditionally required specialized equipment. Multi-sensor packages will provide comprehensive well characterization during routine mechanical interventions connectivity enhancements will enable real-time collaboration between field operations and remote technical support centres. High-speed data transmission will allow expert oversight and guidance for complex interventions, improving success rates and operational safety while reducing onsite personnel requirements.

## Conclusions

This comprehensive study demonstrates that digital slickline technology delivers transformational improvements in well intervention efficiency, cost-effectiveness, and operational safety. The 25% to 35% reduction in total intervention time, combined with 20% operational cost savings and 40% non-productive time reduction, establishes digital slickline as a compelling alternative to conventional methods. The technology's ability to integrate mechanical intervention capabilities with real-time data acquisition eliminates traditional operational processes and enables streamlined intervention programs, single system deployment, reduces equipment requirements, personnel needs, and logistical complexity while improving operational outcomes across all measured parameters. Economic benefits extend beyond direct cost reductions to encompass broader value creation through accelerated operations, reduced production deferrals, and enhanced operational reliability. The technology's proven performance across diverse well conditions and intervention types demonstrates universal applicability and scalable benefits. Safety improvements through real-time monitoring capabilities and reduced personnel exposure establish digital slickline as a valuable tool for enhancing operational safety standards. The technology contribution to risk reduction and incident prevention supports broader industry safety objectives while improving operational outcomes. The successful deployment of digital slickline technology across 25 well interventions validates its reliability and effectiveness under field conditions. Performance consistency and system robustness confirm the technology readiness for widespread industry adoption and integration into standard intervention practices. Future development trajectories promise continued capability expansion and performance improvement, ensuring digital slickline technology remains at the forefront of well intervention innovation. The technology represents a fundamental shift toward intelligent intervention systems that optimize performance while enhancing safety and reducing costs.

## References

Abdullah Alghamdi, Rakan Yateem, Rashed Rajeh, Mohammad Kadem 2018. Real-Time Slickline Production Logging: Technology Review. Presented at the Abu Dhabi International Petroleum Exhibition & Conference, Abu Dhabi, UAE, November 2018. SPE-193288-MS https://doi.org/10.2118/193288-MS

J. Segura,V. V. Tran, J. Meirkhan, L. H. Rustad, V. Varveropoulos 2023. Autonomous Slickline and Wireline Conveyance Improves Performance of Offshore Interventions. Presented at the SPE Offshore Europe Conference & Exhibition, Aberdeen, Scotland, UK, September 2023. SPE-215586-MS https://doi.org/10.2118/215586-MS

Sarunya Jongnarungsin, Feras Abu Jafar, Abdulla Bu Ali, Ajana Laoroongroj, Ahmed El Hariry, Toon Puttisounthorn 2017. Digital Slickline: One Stop Shop for Well Intervention. Presented at the Abu Dhabi International Petroleum Exhibition & Conference, Abu Dhabi, UAE, November 2017. SPE188886-MS https://doi.org/10.2118/188886-MS

Baker Hughes. 2023. "Digital Slickline Technology Overview." Technical Documentation, Baker Hughes Company. Digital slickline | Baker Hughes

Downloaded from https://onepetro.org/SPEADIP/25ADIP/conference/25ADIP by Dmitry Kruglov on 09 November 2025

Paradigm Technology Services. 2021. "Slick-E-Line System Technical Specifications." *Product Documentation*, Paradigm Group B.V. Slick-E-Line® – Paradigm Group

Downloaded from https://onepetro.org/SPEADIP/25ADIP/conference/25ADIP by Dmitry Kruglov on 09 November 2025